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RESEARCH ARTICLE

A history of prenominal passive participles in English: from resultatives to eventives

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Abstract

This article aims to explain how passive participles used as prenominal modifiers developed their eventive nature throughout the history of English. It is argued that prenominal participles first expressed stative result states in Old English (OE) and came to express perfect result states later on. The locus of required resultativity in participles was the inner aspect head in OE, while in Early Middle English (EME), it shifted to the outer aspect head. This shift was triggered by the loss of OE aspectual prefixes, which generally functioned to perfectivize or transitivize the verb by affecting its (internal) argument and assigning a change-of-state meaning to the verb. This shift rendered participial formation to be less constrained, as a result of which, it became possible for prenominal participles to express perfect resultative meanings, which in turn gave rise to their eventive meanings.

Keywords: passive participle; resultative; eventive; perfect; locus of resultativity

1. Introduction

While there has been considerable discussion on the resultative and stative natures of prenominal passive participles (henceforth, participles) in English, less attention has been paid to their eventive nature. Especially how their eventive nature developed from their resultative and stative natures, which is my primary concern in this article, has not been explicated in the literature.

The most effective diagnostic feature of the eventive nature of prenominal participles is that they can be freely modified by certain groups of adverbs. The first group may include temporal adverbs, such as *previously*, *then* and *once*, which normally reject the modification of currently holding result states.[1]

[1] For example, *previously* is bad with *gone*, which produces a perfect-of-result meaning.
(i) He has previously been to Italy.
(ii) #He has previously gone to Italy (several times in the past few years).

(1) So I have one of the previously mentioned things. (COCA [WEB: 2012])[2]

(2) The then accepted interpretation of the artistic achievements of …
(COCA [ACAD: 1997])

(3) … the Marine band in the once hated red coats, … (COCA [WEB: 2012])

The second group includes adverbs of frequency, such as *frequently* and *often*, which normally contribute to composing habitual states but do not modify result states.

(4) … answers to frequently asked questions about donating blood can …
(COCA [NEWS: 2014])

(5) The most often cited reason why minority-owned firms don't get …
(COCA [MAG: 1992])

The third group includes adverbs of manner such as *quickly* and *noisily*, which modify only events and are therefore out in the perfect of result.[3]

(6) …, after a quickly called meeting to coordinate a response, …
(COCA [NEWS: 2009])

(7) …ladies in satin and ermine carefully descended the noisily flipped down footrests, …
(COCA [FIC: 2007])

The temporal adverb *recently* is good in the perfect of result, but when it co-occurs with a prenominal participle, it can modify either a result state or a prior event (Sleeman 2011).[4] When a prenominal participle is resultative, it should be able to appear after copulas such as *remain*, *seem*, *look* and *appear*, which normally select state-denoting predicates, as indicated by the following data.

(8) (a) the carefully opened package
(b) The package remained carefully opened. (Sleeman 2011: 1572)

However, unlike *carefully opened*, which is resultative, *recently opened* fails to occur after such copulas. As Sleeman (2011) notes, this is because *recently opened* is eventive rather than resultative.

(9) (a) the recently opened door
(b) *The door remained recently opened. (Sleeman 2011: 1572)

The eventive nature of prenominal participles is further indicated by the following data. Atelic verbs, such as *talk about*, *pay*, *take*, *shout*, *whisper* and *look after*, do not lexically encode a result meaning; therefore, their participles are neither resultatives nor statives but are good in the prenominal position.

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[2] Examples in (1)–(7), (21)–(23a) and (31)–(32) are from the *Corpus of Contemporary American English* (COCA; Davies 2008–).

[4] See Mittwoch (2014: 29), who provides the following examples.
(i) Jane has translated the poem #quickly/literally.
(ii) They have sealed the door #noisily/hermetically.

[4] As noted by Sleeman (2011), prenominal position is not helpful in distinguishing different interpretations of participles. Many researchers (Embick 2004 and others) have tried to identify the distinct types of participles, using prenominal position as a diagnostic of non-eventivity. It is, however, is too crude to claim that prenominal participles are not Eventives.

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(10) (a) much talked about new show (Sadler & Arnold 1994: 190)
(b) a paid physician (Ackerman & Goldberg 1996: 23)
(c) the most recently taken photos (Lundquist 2013: 13)

(11) Strike could hear voices through the gloom, shouted instructions and the growl and beep of reversing lorries unloading the carcasses. (Albrespit 2020: 10)

(12) The night before, they had received whispered instructions to meet later. (Albrespit 2020: 10)

(13) A child who has been in the care of their local authority for more than 24 hours is known as a looked after child. (Albrespit 2020: 13)

However, Old English (OE) did not have eventive participles as prenominal modifiers, which became available in Early Middle English (EME) onwards. The following sections provide detailed discussion on this, with a focus on a formal analysis of the diachrony of the participles. Section 2 clarifies the interpretations and structural characteristics of prenominal resultative participles (section 2.1) and eventive participles (section 2.2). It is argued that the inner aspect head is the locus of stative resultativity and the outer aspect head is the locus of perfect resultativity and various current-relevance properties. Section 3, after presenting three lines of reasoning to show that OE prenominal participles were stative resultatives (section 3.1), elaborates on how the locus shift in resultativity took place in ME (section 3.2) and how it gave rise to the emergence of eventive participles (section 3.3). It is shown that the shift was triggered by the loss of aspectual prefixes and opened the possibility for more dynamic meanings to be expressed by prenominal participles. Section 4 clarifies that the perfect resultative nature developed in participles in the *have* periphrasis in addition to prenominal participles, thereby demonstrating that the rise of the perfect resultative sense as well as the eventive sense is closely related to aspectual changes the participle underwent on its own. Section 5 concludes this article.

2. Interpretation and structure of prenominal participles

2.1. *Resultative participles*

A given participle, resultative or eventive, can always find a corresponding full clause with respect to its tense–aspect interpretation. For resultatives, there are two types of clause, *be*$_{cop}$ *done* and *have done*. For example, *the broken car* can have either stative or perfect resultative interpretations because it finds canonical structures corresponding to these interpretations. Note that in (14b), *be* is the copula *be*, which normally selects adjectives and nouns, rather than the passive auxiliary, and (14c) exemplifies the *have done* structure with a perfect-of-result meaning, not with an experiential or universal perfect meaning.

(14) (a) the broken car
(b) The car is broken. (Stative resultative)
(c) The car has (been) broken (by someone). (Perfect resultative)

To give a straightforward description in an informal style, for stative resultatives, the result state is foregrounded and the action has faded into the background, and vice versa for perfect resultatives. Put it another way, stative resultatives 'express a state implying a previous event (action or process) it has resulted from' (Nedjalkov 2001: 28), and perfect resultatives denote 'a past action with current relevance' (Bybee *et al.* 1994: 61).

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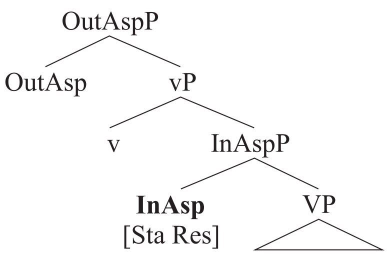
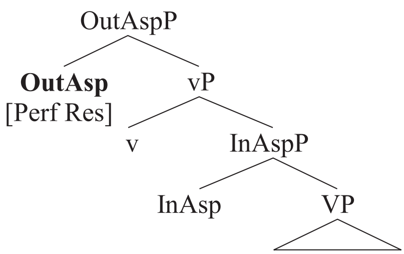
(15) (a) Stative resultative: < event, **state** >
(b) Perfect resultative: < **event**, state >

Given that stative resultatives are lexically restricted (Bybee *et al.* 1994: 66–7; Nedjalkov 2001: 930), requiring that the verb encode a result state or affect its (logical) object, it can be argued that stative resultativity is the inner aspectual property of the verb. In contrast, perfect resultatives do not have constraints on the lexical type of the base verb; both change-of-state (COS) and non-COS verbs can serve as input to perfect resultatives, for example, *the broken car* and *already massaged patients.* This implies that perfect resultativity is an outer aspectual property. This leads to the conjecture that the loci of the two types of resultativity are aspectual heads, Inner Asp and Outer Asp,[5] respectively, which are located below and above the verbalizing head v, respectively, as shown below.[6]

(16)



(17)



A notable difference between these two structures is that in (16), OutAsp is inert with respect to resultativity in that the event associated with the result state is at best implied by the participle but not denoted by it, which is not true of OutAsp in (17).[7] From this it follows that resultativity arises in InAsp, not OutAsP, in (16) and that stative resultatives are derived adjectives, as opposed to simple adjectives. As such, OutAspP in (16) is selected by the auxiliary *be*, not *have.* By contrast, resultativity arises in OutAsp in (17). Alternatively speaking, OutAsp is the locus of resultativity in (17). That is, when interpreted as perfect resultatives, participles are verbs that may be selected by the perfect auxiliary *have* (in a clause-level context). In this case, OutAsp is also the locus of the temporal-aspectual

[5] 'Inner vs. outer aspect' here (cf. Travis 2010) corresponds to the traditional 'lexical vs. grammatical aspect' or 'situation vs. viewpoint aspect' (cf. Smith 1991: 3).

[6] A similar analysis is found in Embick (2004), who postulates a [FIENT] feature on v for stative participles and an Asp head for resultatives. Embick (2004), however, seems to be not concerned with perfect–stative distinction of resultatives.

[7] OutAsp can be characterized as a stativizing head, which contributes a stative semantics to the participle. That is, it encloses the eventuality expressed by vP and produces stativity. In this sense, [Sta], for stativity, is in fact associated with OutAsp, being separate from [Res], for resultativity, in (16).

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properties of the participle. Tense is instantiated and morphologically realized when Out-AspP is selected by *have*, while it remains unrealized in prenominal position, without an overt tense marker.

2.2. *Eventive participles*

According to their temporal-aspectual properties, eventives can be divided into three types. In the following sentences, the event does not necessarily entail a result state holding at the speech time, constituting an experience for the entity denoted by the object. The perfect in such cases is thus characterized as 'experiential perfect'.

(18) Has John ever broken his leg? (Smith 1991: 260)

(19) Someone has locked the door. It happened before I arrived. (Mittwoch 2008: 335)

Importantly, the temporal-aspectual interpretations of the participles in the following list correspond to those of the canonical experiential perfects listed above.[8] Let us call such participles 'experiential eventives'.

(20) Many of you here tonight were once lost children... (COCA [MOV: 1991])

(21) ... an assessment based on his previously broken nose, ... (COCA [NEWS: 2004])

(22) ... women could be loaded into the lifeboat through the previously locked windows ...[9] (COCA [FIC: 1991])

There are participles that qualify neither as resultatives nor as experiential eventives. Instead, they correspond to canonical, simple past passives with respect to their temporal-aspectual properties. Let us call such participles 'simple-past eventives'.

For example, *buy* denotes a buying event that is not repetitive with respect to a single entity; it is not appropriate to say that you bought a house many times. Thus, an experiential eventive reading is excluded in (23).[10] A resultative reading is also excluded in that *bought*, with or without *newly*, is completely out in the post-auxiliary position.[11]

(23) (a) ... repairs or environmental hazards in a newly bought house, ... (COCA [ACAD: 2005])

(b) *The house remains newly bought.

[8] See also examples in (1)–(2) and discussion of them.

[9] These participles can, of course, also be resultatives evaluated in the past time, meaning that the result state existed before, but not now. This, however, does not change the fact that there is no result state lasting to the present in such examples. *A newly bought house* can certainly be paraphrased by *a house that has newly been bought* but the event denoted by the participial phrase in the latter is interpreted like 'hot news'. Hot-news/Recent-past events, although they are expressed by the perfect in English, are neither experiential nor resultative, as discussed by Schwenter (1994). Schwenter (1994: 1007) argues that the hot news perfect represents the endpoint of the perfect category, implying that it is more like the simple past than the other three perfects (resultative, experiential, and universal). On the other hand, *a newly bought house* can also be paraphrased by *a house that was newly bought*, which is a case of the simple past. All this points to the simple-past interpretation of the prenominal *newly bought.*

[10] Experiential is subject to a condition that the event has to be repeatable (Mittwoch 2008: 327 and sources therein).

[11] Note that simple-past eventives as well as canonical simple pasts do not evaluate the extended-now (and/or current-relevance) property.

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Thus, *bought* in (23) is interpreted as instantiating the simple past, just as in (24), in which the prenominal participles and their canonical counterparts are both simple-past tensed.

(24) the recently made headway – all that headway was made in a day. (Lundquist 2013: 13)

(25) the most recently taken photos – these photos were taken recently. (Lundquist 2013: 13)

For example, without adverbial modification, consider *shouted* and *whispered* in (12)–(13), repeated below as (26)–(27). They are not resultatives given that they are out in any post-auxiliary position.

(26) Strike could hear voices through the gloom, shouted instructions and the growl and beep of reversing lorries unloading the carcasses.

(27) The night before, they had received whispered instructions to meet later.

(28) *The instructions are/seem/remain shouted/whispered.

Shouted and *whispered* in such cases do not instantiate repeatability and do not evaluate the extended-now property, indicating that they do not qualify as experiential eventives either.

The third type is habitual eventive. For habitual eventives, the participle denotes an event-set with the time at which an event occurs, which is left unspecific.[12] By 'unspecific', I mean that the event time is specified neither as past nor as present and nor as future. The unrealized tense is understood as present, subsuming future, by default. But when there is a clear indicator, it can also be past. This situation corresponds to the canonical habitual sentences described below:

(29) Present (by default) habitual:
My dad always yells instead of talking, and he has for years. He yells at me for doing stuff ... and not doing stuff. (Binnick 2005: 340)

(30) Past habitual:
The tiger ate many fish live and whole .... (Binnick 2005: 340)

As indicated by the adverbs of frequency, the participles, namely, *asked* and *cited*, in the following sentences denote events that last (or lasted or have lasted) for an extended period of time or take place (took place or have taken place) repeatedly; they do not denote result states that currently hold or previously held.

(31) ... answers to frequently asked questions about donating blood can ... (COCA [NEWS: 2014])

(32) The most often cited reason why minority-owned firms don't get ... (COCA [MAG: 1992])

Regarding the formation of eventives, the following structure, which is essentially the same as that of perfect resultatives, is assumed.

[12] By 'event-set', it is meant that not a single event but rather a series of individual events constitute a habitual situation.

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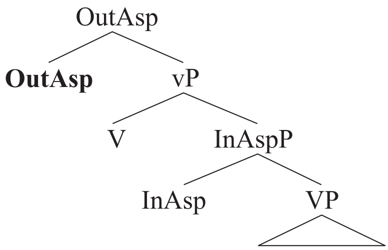
(33)



However, for eventives, OutAsp does not take care of a result state. It instead functions to connect the eventuality to a situation holding at the evaluation time. OutAsp is thus the locus of the extended-now property in the case of experiential eventives. Note that the extended-now property is what connects a past event with a present situation.

In the case of simple-past eventives, OutAsp is the locus of past-shifting that arises with the event denoted by the participle. Consider (13), repeated below as (34).

(34) A child who has been in the care of their local authority for more than 24 hours is known as a looked after child.

Looked after here is a simple-past eventive rather than an experiential eventive.[13] A looked after child is one who is uninterruptedly looked after in the past 24 hours, but not one who has a past experience of being looked after for 24 hours. Note that the perfect *has been in the care of their local authority for more than 24 hours* instantiates a universal perfect rather than an experiential one. Specifically, *looked after* denotes an event, durative or not, that is located in the past with respect to the evaluation time,[14] and there is no interval between the ending point of the event (looking after for 24 hours) and the beginning of the child's being characterized as 'looked after'; the ending and beginning points overlap in some sense. Put differently, the child's becoming a looked after child uninterruptedly follows the preceding event's ending. *Looked after* is thus used to characterize a child to whom a looking-after event happened, with the emphasis simply on the pastness of the event but not on the continuative, universal property of it. What is evaluated at the evaluation time is (the pastness of) the event but not its continuative property.

For another example, when a photo was recently taken, the event serves to characterize the photo, hence 'a recently taken photo'. Thus, 'recently taken' is a situation of the photo holding at the evaluation time. This is how a past event is connected to a present situation. We could speak of this property as a past-shifting property. That is, there is a shift from a past event to a present situation. As is the case with the extended-now property, the past-shifting property is essentially a temporal-aspectual but not a lexical semantic property. This said, this property arises in a domain above vP.

Hallmann (2021) states that past-to-present shifting is a characteristic of resultatives and is restricted to cases in which the base verb is eventive per se. For example, while *a repaired motorcycle* denotes ('implies' in his word) a past repairing event, a well-known problem does

[13] It does not seem that *looked after* qualifies as a Resultative, either. Strictly speaking, a child's being characterized as 'looked after' is not a state resulting from the looking after event. If it is a result at all, it is a result of characterizing as such rather than of the looking after event itself. The *still* test and the *remain* test also rule out *looked after* as a Resultative.

(i) *That child is still looked after. (fine under a habitual reading)

(ii) *That child remains looked after.

[14] It is the evaluation time rather than the reference time that is relevant here.

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not denote ('imply' in his word) a past knowing state. Hallmann (2021) is not concerned with the past-shifting property of eventives, however. But it is not difficult to see that past-shifting occurs exactly in the *looked after* and *recently taken* examples, which exemplify the simple-past eventive.

A short note is in order here. Past-shifting does not necessarily involve the speech time, although the speech time is often the evaluation time by default (Hatav 2012: 612). It can also involve a past time if the evaluation of the situation is located prior to the speech time. This means that past-shifting may be specified as past-to-evaluation-time shifting, with the evaluation time being either present or past. Note that a past-to-present situation will become a past situation over time. However, the participle is used to describe an entity whose situation is located at the evaluation time or within a time interval containing the evaluation time. Notably, the evaluation time is present in the *looked-after* example and past in the *shouted* and *whispered* examples.

For habituals, OutAsp is the locus of the habitual aspectual property, which is state-related in the sense that it represents a series of individual events that, as a whole, make up a state stretching back into the past and forward into the future (Leech 1971: 5). Scopally, an individual event or a set of individual events are embedded by a habitual situation. Thus, vP and InAspP do not concern a habitual situation but OutAspP does.

Summarizing the above discussion, it is clear that for eventives, OutAsp is a head that takes care of a particular evaluation-time relevant property of the situation denoted by the participle. Broadly speaking, this property is simply current-relevance.[15] It may be an extended-now property, a past-shifting property, or a habitual aspectual property.

3. From resultatives to eventives

Ritz (2012: 884) summarizes from Bybee *et al.* (1994), Dahl & Hedin (2000) and Nicolle (2012) that 'Perfects often take their origins in resultative constructions, with a shift of meaning from current result to current relevance. This shift is characterized by an expansion in the types of verbs used, from exclusively telic verbs in resultative constructions, to both telic and atelic verbs in perfect constructions.' Lindstedt notes that the resultative usually changes into the perfect (2000: 368), which in turn undergoes a gradual transition from the aspectual to the temporal domain (2000: 369), and historically the experiential meaning derives from the current-relevance meaning of the perfect (2000: 370). In what follows, it is shown that adjectival participles in English followed a similar course of development, given the parallelism of temporal-aspectual properties between prenominal participles and post-auxiliary participles.

3.1. Prenominal participles as stative resultatives in OE

While participles in the post-auxiliary position have been widely discussed, those in the prenominal position have received less attention. In this subsection, it is explicated that prenominal participles are stative resultatives in OE and provide three reasons for this claim as follows:[16] (i) the obligatoriness of the aspectual prefix, (ii) restriction on adverbial modification and (iii) the relationship between inflections and word order.

[15] What has been known as current relevance in the literature is a property of the present perfect. In this article, however, the term 'current relevance' is used in a broader sense; any construction denoting an eventuality that has relevance in any pattern with a current situation can be said to have a current-relevance property.

[16] Recall that the periphrases with both *have* and *be* are generally assumed to have started out (pre-OE) as statives built around resultative participles (McFadden & Alexiadou 2010: 415).

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It is well documented that past participles in OE are often marked with certain prefixes expressing the perfective aspect (cf. Brinton 1988: 202ff.; Elenbaas 2007: ch. 4; McFadden 2015, 2023, etc.), which are listed below.

(35) *a-, be-, for-, forþ-, ful- (full-), ge-, of-, ofer-, to-, þruh-, up-, ut-, ymb-.*
(Brinton 1988: 202–3)

Elenbaas (2007: 114) states that 'OE prefixes have a range of meanings, but at the core, they share common semantics. The meanings of the prefixes are invariably abstract, and the prefixes typically denote an end state and express the total affectedness of the object.' She also states that the core denotation of the prefixes is a COS meaning (2007: 118). Some of them, such as *be-*, *for-*, *ge-* and *ut-*, have transitivizing effects, he notes (2007: 116–18, 158).

Importantly, most of prenominal participles in OE (94%), according to corpus investigation, were marked with these prefixes in (35), among which, *a-*, *be-*, *for-* and *ge-* were the most frequently used, as shown in table 1.

This suffices to show that to be marked with a prefix is a condition for a verb to derive a participle as a prenominal modifier in OE. As shown in the table, the prefix *ge-* is most frequently used. The same holds true for participles in post-auxiliary position. McFadden (2015) observes that *ge-* can appear on any form of the verb, but it was most frequent with past participles (89%), as shown in table 2.

Table 1. Distribution of prefixes of prenominal participles in OE[17]

Prefixed	*a-*	62	92 (%)
	be-	24	
	for-	28	
	ge-	93	
	others	46	
Non-prefixed		21	8 (%)

Table 2. Distribution of *ge*-participles, according to verb type (McFadden 2015: 22)

Form	with ge-	without ge-	with *ge-* (%)
Present participle	107	1,493	6.7
to infinitive	430	2,177	16.5
Finite	23,723	102,434	18.8
Bare infinite	4,329	11,188	27.9
Imperative	2,273	5,468	29.4
Past participle	11,505	1,484	89.0

[17] Multiple occurrences of the same participle with a particular prefix count as a single occurrence in the table.

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Table 3. Distribution of *ge*-participles, according to auxiliary (adapted from McFadden 2015: 38)

Form	with *ge-*	without *ge-*	with *ge-* (%)
Participle with *be*	861	10	99
Participle with *have*	125	4	97

Moreover, the frequency of the prefix increases (99% and 97%) when the participle is used with an auxiliary, as shown in table 3.

The data in tables 2 and 3 imply that participles with *ge-* in clauses indeed expressed stative result (his 'resultative' as opposed to his 'perfect') states, as concluded by McFadden (2015: 38). Considering the data in table 1 in parallel with this, it is reasonable to conclude that prenominal participles express stative result states in OE. It is suggested here that as far as post-auxiliary participles are stative resultatives, the corresponding prenominal participles are hardly perfect resultatives in principle, in the sense that eventive interpretations are more restricted in prenominal attributive positions than in post-auxiliary positions. Recall that stative resultatives point to the result state while perfect resultatives point to the action itself (cf. Bybee *et al.* 1994: 64 and §2.1).

Another fact that supports the claim that prenominal participles are stative resultatives in OE is that no prenominal participles were attested on adverbs that are often used to modify events, but not result states, according to my corpus investigation.

(36) (a) the recently opened door, the previously mentioned things
(b) a quickly called meeting, the noisily flipped down footrests

Adverbs modifying prenominal participles were mostly degree adverbs or intensifiers (cf. Visser 1963–73: 1232). Note that degree adverbs and intensifiers generally have nothing to do with events but with states.

(37) a well-established tradition, a badly damaged house, a very respected scholar

Visser (1963–73: 1235–6) presents a list of prenominal participles modified by a preceding *-ly* adverb. Among the examples, most of which seem to be manner adverbs, the earliest one was attested around 1400: … *with grynly grownden gare* 'a grimly grown gore'. Visser (1963–73: 1233–4) also presents a list of the *our dear bought victory* type of prenominal participles, where the word preceding the participle is also an adverb. Importantly, however, the earliest example in Visser's list was attested in *c.*1205: *hæheste iborne mon* 'highest-born man'.

All these facts concerning adverbial modification put us in a position to argue that prenominal participles, lacking the perfect resultative sense, expressed stative result states in OE, and this is why adverbial modification was restricted for them to a certain extent.

Another fact supports the claim that prenominal participles are stative resultatives in OE. Mitchell (1985: §710) writes, 'Caro (1896: 404–6) … gives figures which confirm the reasonable expectation that the object-participle order produces a higher percentage of inflected forms (twenty-seven per cent) than the participle-object order (fifteen per cent).' Although it has been observed that inflected and uninflected participles can coexist in the same clause (Mitchell 1985: §711; Traugott 1992: 191), the difference in frequency between inflected and uninflected participles with respect to word order itself suffices to indicate

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that there must have been a certain relationship between inflection and the function and meaning of the participle.[18] This can be illustrated as in (38) and (39).

(38) (a) *have*-object-participle: participle inflected mostly
(b) *have*-participle-object: participle uninflected mostly

(39) (a) America [$_{VP}$ [$_{V}$ has][[$_{NP}$ a role] [$_{A}$ found]]].
(b) America [$_{VP}$ [has [[$_{V}$ found]] [$_{NP}$ a role]].

The data in the present investigation, which were obtained from *The York–Toronto–Helsinki Parsed Corpus of Old English Prose* (YCOE) (Taylor *et al.* 2003) and *The Penn–Helsinki Parsed Corpus of Middle English*, second edition (PPCME2) (Kroch & Taylor 2000), as summarized in table 3, more clearly show the relationship between the presence/absence of inflections in the participle and word order.[19] As shown in table 4, clauses with the *have*-object-participle order and those with the *have*-participle-object order both existed in OE, while only the latter did in Middle English (ME). Importantly, as shown in table 5, participles were inflected *only* in clauses with the *have*-object-participle order in OE.

An OE example of an inflected participle with the *have*-object-participle order is given in (40), an OE example of an uninflected participle with reverse order is given in (41), and an ME example with the *have*-participle-object order is given in (42). Notably, the examples in (41) and (42), with the uninflected participles, already have a Present-Day English (PDE) order. In contrast, the example in (40) shows that the participle, following the object and being inflected, expresses a stative result state and can be categorized as an adjective in the traditional sense.[20]

Table 4. Distribution of the *have* periphrasis in OE and ME[21]

	O1	O2	O3	O4	M1	M2	M3	M4
Have-object-participle	1	33	68	17	0	0	0	0
Have-participle-object	0	33	67	14	237	124	784	940

Table 5. Distribution of inflected participles in the *have* periphrasis in OE

	O1	O2	O3	O4	M1	M2	M3	M4
Have-object-participle	0	9	24	7	/	/	/	/
Have-participle-object	0	0	0	0	0	0	0	0

[18] Mitchell (1985: §711), Traugott (1992: 191) and Denison (1993: 341), for example, seem to also take this stance.

[19] I restricted my search of YCOE and PPCME2 to auxiliary(*have*)-initial clauses in which an accusative object immediately precedes the participle and those in which the participle precedes an accusative object.

[20] The translation of the OE examples into PDE in this article is based on *An Anglo-Saxon Dictionary* (Bosworth & Toller 1898) and the translation of ME examples is based on *A Middle-English Dictionary: Containing Words Used by English Writers from the Twelfth to the Fifteenth Century* (Stratmann & Bradley 1891).

[21] The periodization of the history of English in the corpora is as follows: O1 (–850), O2 (850–950), O3 (950–1050), O4 (1050–1150), M1 (1150–1250), M2 (1250–1350), M3 (1350–1420), M4 (1420–1500), E1 (1500–1569), E2 (1570–1639), E3 (1640–1710), L1 (1700–70), L2 (1770–1840), L3 (1840–1900). In tables 4 and 5, O1 and O2, O3 and O4, M1 and M2, and M3 and M4 are combined as EOE, LOE, EME and LME, respectively. The term 'Early English' in this article is used as a cover term for OE (to 1150), ME (from 1150 to 1500) and EModE (from 1500 to 1710).

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(40) Ðu hæfst me nu **manega** **bysna** **gereihte**,
you have me now many examples:F.PL.ACC explained:F.PL.ACC
Lit: 'You now have many examples explained to me'
(cosolilo,Solil_3:66.26.926: O2)

(41) Nu hæbbe we **awriten** **þære** **Asian** **suþdæl.**
now have we written the Asian south (land)
'Now we have figured out the Asian's south land.'
(coorosiu,Or_1:1.11.25.167: O2)

(42) For treuly thou hast **chose** **the** **best** **partye.**
for truly you have chosen the best party
'For you have truly chosen the best party' (CMAELR4,20.568: ME)

Elsness (1997: 261) notes that inflected participles are found more in the *be* periphrasis than in the *have* periphrasis. This is consistent with the data in table 5 in that only the *be* periphrasis and the *have* periphrasis with the *have*-object-participle order allow stative resultative readings. Because the participle therein is interpreted as stative, they are more adjective-like than those in the *have* periphrasis with the *have*-participle-object order, in which the participle has a stronger verbal sense.

It then follows that the inflections of the participles must have been related to their stative resultative meaning, which is a stance taken by many authors, including Mitchell (1985: §711), Traugott (1992: 191) and Denison (1993: 341), as opposed to Brinton (1994: 140) and Wischer (2004: 249).

Given these three lines of reasoning, that is, each concerning (i) the obligatoriness of aspectual prefix, (ii) restriction on adverbial modification and (iii) the relationship between inflections and word order, it can now be concluded that prenominal participles in OE are indeed stative resultatives. In the following subsections, it is shown that prenominal and post-auxiliary participles ceased being restricted to stative resultatives and developed a perfect resultative sense after OE.

3.2. The shift of resultativity locus led to perfect resultativity

As has been well documented in studies including Jespersen (1931: 29–31), Visser (1963–73: 2043–4, 2189ff.), Traugott (1972: 144–6; 1992: 191–3), Mitchell (1985: §728–33), Brinton (1988: 99–102), Parsons (1990: 242–5), Bybee *et al.* (1994: 68ff.), Haspelmath (1994: 161ff.), Carey (1996: 35–40), McFadden & Alexiadou (2006, 2010) and so on, participles in the *have* periphrasis had a resultative meaning in OE mostly when they had the object–participle order, as illustrated by PDE examples in (43). In contrast, participles with the participle–object order, like those in (44), had only a perfect meaning. Relevant representations are given in (45).

(43) Resultative:
(a) I have my work done. (Visser 1963–73: 2189)
(b) I have the house built. (Brinton 1988: 100)
(c) I have him bound. (Parsons 1990: 244)

(44) Perfect:
(a) I have done my work. (Visser 1963–73: 2189)
(b) I have built the house. (Brinton 1988: 100)
(c) I have bound him. (Parsons 1990: 244)

(45) (a) America [$_{VP}$ [has [[$_{V}$ found]] [$_{NP}$ a role]]. (Denison 1993: 340)
(b) America [$_{VP}$ [$_{V}$ has][[$_{NP}$ a role] [$_{A}$ found]]]. (Denison 1993: 340)

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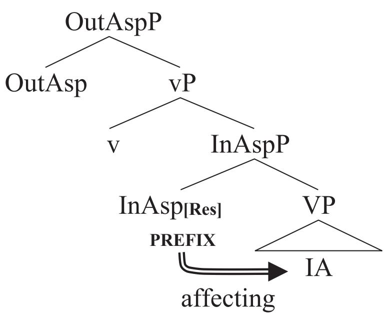
Two points here have been agreed upon by authors. First, the perfect resultative meaning became available or more salient throughout Late OE (LOE) and EME.[22] Second, the types of verbs selected in perfect resultatives have expanded toward PDE; put differently, the base verb of past participles has become less restricted to certain types[23] and more additional meanings, such as experiential and universal as subtypes of the perfect, have become available.[24]

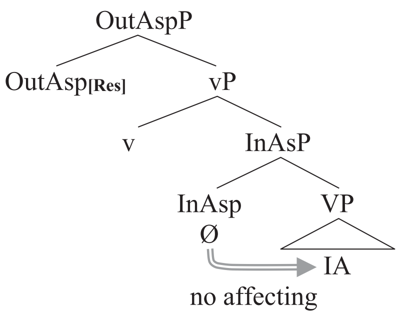
However, most, if not all, of these authors have focused on the '*have* + participle' constellation, with less attention having been paid to changes the participle underwent on its own. As we will see in what follows, aspectual changes in the participle are more significant for the rise of the perfect resultative sense of prenominal participles.[25]

As discussed in section 2.1, InAsp is responsible for stative resultativity and OutAsp for perfect resultativity. Given that normally only verbs denoting actions that affect the entity (e.g. COS verbs) serve as input to stative resultatives, it is predicted that the affectedness has been overtly marked on the participle in OE, which is a highly inflected language. This prediction is borne out and supported by the fact discussed in the previous subsection that OE participles are always marked with aspectual prefixes, which are derivational rather than inflectional (cf. Visser 1963–73: 1223; McFadden 2015: 17, 41). However, this was not true for participles in EME onwards because affixes inflected on adjectival elements began to disappear. Each pattern is represented as follows:

(46) The structure of Stative Resultatives in OE:



(47) The structure of Perfect Resultatives in EME onwards:



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[22] See Traugott (1972: 93), Mitchell (1985: §728–33), Carey (1996: 34) and McFadden & Alexiadou (2006: 277; 2010: 392), for example. Those who disagree with these authors include Brinton (1988: 100) and Wischer (2004: 249). It seems, however, that disagreement among authors exists over the analysis of individual examples, not over the construction in general (cf. Ringe & Taylor 2014: 437).

[23] See Bybee *et al.* (1994: 69ff.), McFadden & Alexiadou (2010: 392), for example.

[24] See Carey (1996: 35), McFadden & Alexiadou (2010: 401–2). Lee (2003), however, claims that experiential and universal meanings were already available for OE *have* periphrasis.

[25] See section 4 for discussion on how aspectual changes in the participle are correlated with the *have* and *be* periphrases.

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In this sense, the formation of the participles should be like (48a), not like (48b). The prefix is closer to the verb nucleus than to the suffix *-en*, which is inflectional in nature.

(48) (a) [*-en* [*ge-* [V]]][26]
(b) [*-ge* [*-en* [V]]]

The proposal herein is partly inspired by van Gelderen (2011), who takes the aspectual prefix *ge-* as located in InAsp, where it affects the internal argument (theme, in van Gelderen 2011: 110).[27] Verbs that affect their internal argument are often those that denote COS (cf. Tenny 1987: 70). Thus, it is reasonable to assume that OE aspectual prefixes are lexical realizations of InAsp. This is because they typically denote an end-state, express the total affectedness of the internal argument, and assign COS meaning to the verb (Elenbaas 2007: 114) so that the participle is able to express stative result states.

Note that the present analysis is compatible with the fact that OE participles do not take unergative verbs as their inputs, as discussed by McFadden & Alexiadou (2010: 391, 412) among many others. Unergative verbs are ruled out because they do not generally have an internal argument, so that in participial formation, the aspectual prefix has no way of affecting an argument (cf. Haspelmath 1994: 159).[28] Perfect resultativity, however, is located on OutAsp.

Notably, aspectual prefixes were lost in ME, and the required resultativity no longer had overt morphemes; this means that the inner aspect head was not morphologically realized in ME. Another question arises: how would prenominal participles express the required resultativity now? Before answering this question, it must be noted that the required resultativity of prenominal participles was typically marked by aspectual prefixes. After the aspectual prefixes were lost, something would take over the task from the aspectual prefixes. The resultativity required for the prenominal position would need to be marked by some other overt items. The question then becomes: what served as the marker of resultativity? The answer is that the participle ending *-en* did. This change is expressed as follows:

(49) (a) [*-en* (not marker of res.) [*ge-* (marker of res.) [V]]] (OE)

↓

(b) [*-en* (marker of res.) [Ø [V]]] (EME onwards)

Notably, in (49a), resultativity is expressed via *ge-*, which represents the various aspectual prefixes in OE, and stativity is expressed via *-en*, which represents the various strong and weak endings of OE participles. In (49b), *-en* functions to mark resultativity and the completion of events, where a result state is implied but not denoted or marked

[26] This structure is strongly supported by the fact that *ge-* was not attached to noun-based participles, namely, *-ed* adjectives, in OE, as noted by Visser (1963–73: 1223). The noted fact suggests that the prefix was part of the base verb.

[27] Consistent with this is McFadden (2015: 38), who also associates *ge-* with inner aspect (*res*, in his terms) rather than outer aspect.

[28] Van Gelderen (2011: 110) claims that in OE, *ge-* used to function as adding a theme argument to the verb, whereby a valency change took place in the history of certain verbs.

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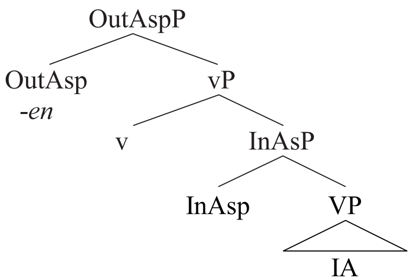
by *-en.* Importantly, the resultativity marked by *-en* is perfect resultativity, not stative resultativity.

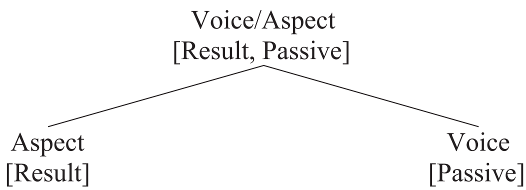
The structural position of this new marker *-en* must now be identified. McFadden (2015: 40–1) locates this morpheme on the head of Asp_{R}P, which is higher than the initiator phrase, *init*P, which is postulated by Ramchand (2008).[29] McFadden's Asp_{R}P corresponds to OutAspP in (46)–(47). Thus, the morpheme *-en* is located on OutAsp, as shown below.[30]

(50) The structural position of *-en*:



This seems plausible at first glance. However, a closer look reveals that the position of the morpheme *-en* may not be in OutAsP because this morpheme can be the realization of passive participles, that is, it might occupy the voice head, which is not represented in (50), given that prenominal participles in OE are typically formed from transitive verbs (cf. Bai 2016). To identify the precise position of the morpheme, we must take a broader view. It is not difficult to see that in participles, especially prenominal ones in OE, voice and aspect cannot be morphologically differentiated. In this respect, Cowper & Hall (2012) claimed that in Early English, voice and aspect were bundled on a single head and subsequently split into separate projections, as shown below.

(51) The separation of voice and aspect



(Cowper & Hall 2012: 131)

If Cowper & Hall's (2012) claim is correct, it is reasonable to say that *-en* is located on OutAsP, which is responsible for both the result and passive effects. When voice and aspect were separated sometime after EME, *-en* came to occupy voice.

Returning to the initial question of what took over the task from OE aspectual prefixes, it could now be concluded that *-en*, the lexical realization of OutAsp (and of voice as well) took over that task. This amounts to saying that the locus of the required resultativity has shifted from InAsp to OutAsp. The representation is now modified as follows:

[29] Based on Ramchand's (2008) verbal composition approach, McFadden (2015) locates *ge-* on the head of a result phrase, *res*P, which roughly corresponds to InAsp in (46) and (47).

[30] This is the same for McFadden & Alexiadou (2010). They, however, ignore the inner aspect.

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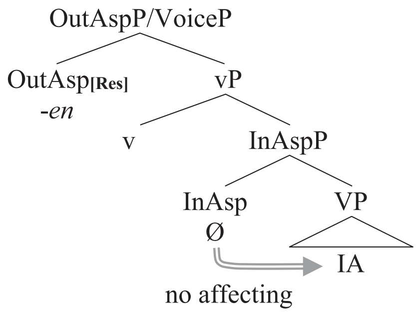
(52) The structure of Perfect Resultatives in EME onwards:



The required resultativity is not restricted to stative resultativity after the shift of resultativity locus. Recall that InAsp is responsible for resultativity by virtue of bearing the aspectual prefix in OE. Because aspectual prefixes were lost in EME, nothing could affect the internal argument nor assign COS meaning to the base verb, leading to the fact that any type of verb could be available as input to the participle in principle. In contrast, OutAsp, which was inert with respect to resultativity in OE, became active and open to the required resultativity after OE.[31] As far as this happened, the perfect resultative sense must have become available because the outer aspect, unlike the inner aspect, is a grammatical functional category rather than a lexical one.

3.3. A consequence: prenominal eventives emerge

The shift of resultativity locus opened the possibility that more dynamic meanings, such as the experiential perfect in addition to perfect-of-result (perfect resultativity here), became possible in prenominal participles. Specifically, in ME onwards, OutAsp was not necessarily a stativizing head; the participle came to also denote an event, where OutAsp, occupied by *-en*, took care of event completion. The speaker presents an event as a bounded whole, where current relevance arises in two patterns. One pattern is 'implied result state' and the other is 'time span', commonly called 'extended-now'. An implied result state is obtained in the case of perfect resultatives and a time interval extends to contain the current time point in the case of experiential perfects. Thus, perfect resultatives and experiential perfects are correlated through the speaker's viewpoint. For example, in *already massaged patients*, the speaker views (or presents) the massaging event as a bounded whole with relevance to a current state, and in *previously massaged patients*, the event is viewed as in existence in a time interval terminating at the current time point. As the speaker's viewpoint naturally connects perfect resultatives and experiential perfects, some functional head serves to be the anchor to bind them together, speaking from the formal syntactic perspective. That anchor is OutAsp. Note that the speaker's viewpoint is associated with OutAsp, not with InAsp, since it is a perspective on the temporal-aspectual nature of the event as a whole, but not on its inherent nature. Eventivity is licensed by both perfect resultatives and experiential perfects. However, only perfect resultatives license stativity by means of implicature, not denotation. Experiential perfects license the extended-now property in addition to eventivity. On the other hand, both a result state and the extended-now property are

[31] Cowper & Hall (2012: 137–8) present a similar structure, suggesting the possibility that OutAsp was inert at first and subsequently became active while InAsp became inert. This assumption seems to be problematic for their analysis of passivals in LModE but not for the present analysis.

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patterns of current relevance, which allows us to say that OutAsp is also a locus of current relevance.[32]

All this said, it is reasonable that once OutAsp gets activated for perfect resultatives, it will call for experiential perfects and give rise to them at a later time, which, I argue, is what happened in the history of prenominal participles. Thus, the eventive nature of eventive participles is a result of the extension of OutAsp's function, which is preceded by the shift of resultativity locus.

This led to the fact that eventive participles with simple-past semantics and habitual semantics have also become available.

(53)	(a) the once lost gun, previously mentioned topics	(Experiential eventive)
	(b) shouted instructions, just announced new visa rules	(Simple-Past eventive)
	(c) frequently asked questions, the most often cited reason	(Habitual eventive)

Recall that the temporal-aspectual properties displayed by these three types of participles are all grammatical properties, not lexical ones. Thus, they are all associated with OutAsp,[33] not with InAsp. Dissociated from InAsp, participles do not restrict the aspectual type of the base verb. Thus, participles of verbs without encoding a COS or result meaning basically do not denote result states but denote completed events.

Note that past-shifting is another pattern of current relevance.[34] Thus, OutAsp, being the locus of current relevance, extends to take care of the shifting of past events to situations holding at the evaluation time.

For perfect resultatives, experiential eventives and simple-past eventives, *-ed* is supposed to be the spell out of OutAsp, implying that the event exists in the past and/or is completed. In contrast, habitual eventives do not necessarily express the pastness or completion of the event. They instead express events that occur regularly or repeatedly. Habitual meaning arises from a generic quantification over time intervals. In this sense, *-ed* spells out (passive) Voice, not OutAsp, as exemplified by *frequently asked questions.* OutAsp thus takes care of generic quantification, which is a notion associated with the outer aspect.

As regards how to formally identify eventives as associated with OutAsp, adverbial modification is arguably the most effective diagnostic and the most important clue to clarifying when they became available or more salient in the history of English. As discussed in section 3.1, there was a restriction on adverbial modification; only degree adverbs and intensifiers were able to modify prenominal participles in OE. The same is true of prenominal adjectives, as noted by Fischer (2001). This implies that the expansion of adverb types may parallel that of verb types in prenominal participles. This is already clear from Visser's (1963–73: 1234–7) data, as also noted in section 3.1. Visser shows that the occurrence of the most canonical types of adverbs and prenominal participles in PDE is not found until EME.[35]

Thus, the emergence of eventives was a later development in the history of prenominal participles, in parallel with that of canonical passives, although they may not have been exactly simultaneous. In other words, the developmental paths of participles in the two contexts have interacted and have been closely related to each other. Recall that the experiential and universal meanings of the perfect followed the development of its

[32] See section 2 for relevant discussion.

[33] Of course, they may be split onto different aspect heads, if we adopt Cinque's (1999) functional hierarchical approach. What is certain is they can never be associated with InAsp.

[34] Current relevance is irrelevant in the case of simple past sentences. It is relevant in the case of simple-past eventive participles because current relevance in any pattern is required and coerced in prenominal attributive position.

[35] The earliest example in his list was attested in *c.*1205 (see section 3.1).

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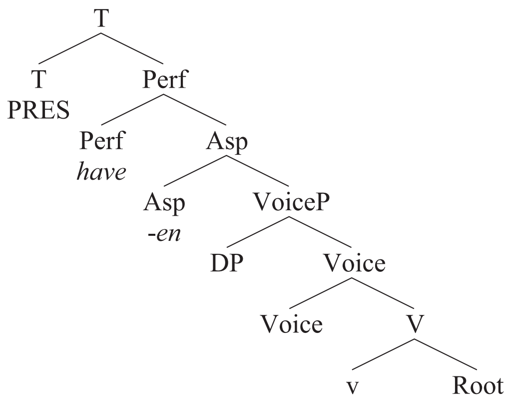
resultative meaning, which took place throughout LME and Early Modern English (EModE). Simple pasts and habituals in a clause-level context, which are distinct from perfects, had a different developmental path, which is beyond the scope of this article.

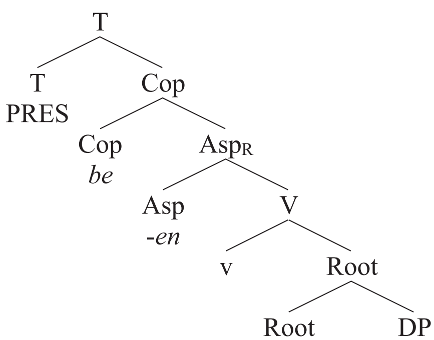
4. Participles in the *have* and *be* periphrases

As mentioned in section 3.2, previous studies have focused on the interaction of participles with the auxiliaries *have* (and *be*). However, no particular concern has been raised about what changes the participle has undergone on its own and how the changes have contributed to the development of the constructions. Recently, some authors have attempted to approach related issues from a formal perspective. One familiar study is McFadden & Alexiadou (2010), who analyze the development of the perfect construction in connection with the auxiliaries *be* and *have*, arguing that *have* has undergone a change in its semantics and syntax, while *be* has not. They conclude that 'the *have* periphrasis involved a clause-level perfect head denoting anteriority, while the *be* periphrasis was a copular construction built around a stative resultative participle' (2010: 421), as diagrammed below.

(54) The structure of the *have* periphrasis (LME (Late Middle English) onwards):



(55) The structure of the *be* periphrasis throughout the history of English:



(McFadden & Alexiadou 2010: 410–11)

The difference between these two structures lies primarily in the fact that (54) involves a perfect head, which, spelled out by *have*, is responsible for creating an extended-now

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interval extending into the past (2010: 411), while (55) involves a copula head, spelled out by *be*, which is nothing more or less than the normal copula that appears in predicate adjectives and nouns (2010: 412). McFadden & Alexiadou (2010: 409ff.) propose that the *have* periphrasis in Late Modern English (LModE) onwards contains material at the clausal-tense-aspect level denoting anteriority to the reference time, developing experiential perfect semantics, and *have* started to spell out the perfect head. In other words, the structure in (54) was not available until LME. According to McFadden & Alexiadou's (2010) analysis, the emergence of a perfect resultative sense for the *have* periphrasis was primarily due to the rise of the perfect head and a change, if any, in the participle selected by *have* was less relevant.

However, McFadden & Alexiadou's (2010) analysis faces some difficulties. The first difficulty can be presented by the question: how did various types of verbs, in particular atelic verbs, in addition to COS verbs, which are basically telic, come to be selected in the *have* periphrasis but not in the *be* periphrasis? According to McFadden & Alexiadou's (2010) analysis, the answer is that the expansion of verb types is simply due to the semantic change of *have.* However, it is not clear under this analysis how a semantic change has taken place with *have* or, to put it another way, what exactly has caused *have* (or the perfect head) to obtain more dynamic senses, such as experiential perfect. With this question remaining open in this analysis, it necessitates a more in-depth consideration and analysis of the factors that have caused the rise of perfect from stative resultatives in general.

Another difficulty with McFadden & Alexiadou's (2010) analysis is that it ignores to some extent the widely accepted view and long held belief that the participle itself has undergone a change from expressing states to expressing actions or, put another way, a change from functioning as an adjective to functioning as a verb (cf. Mitchell 1985: §728; Traugott 1992: 191–3, and many others). According to McFadden & Alexiadou (2010), the difference between stative and perfect resultatives lies only in the outer aspect head, with inner aspect, traditionally called Aktionsart, being ignored.[36] In (55), $\mathrm{Asp_R}$ is intended to capture stative resultativity, but it is the head of outer aspect. However, stative resultativity is associated with inner aspect, not outer aspect. It is the head of inner aspect that takes care of resultativity, which primarily arises with COS verbs or verbs with result-state semantics. If $\mathrm{Asp_R}$ as the head of outer aspect takes care of resultativity as suggested by McFadden & Alexiadou (2010), then the resultativity it produces should be perfect resultativity, not stative resultativity. Unlike stative resultativity, perfect resultativity is not necessarily brought about by an event involving a change of state, as in *The patients have been massaged today* and *already massaged patients.* For perfect resultativity, the lexical aspectual type of the base verb is unrestricted, which is not true for stative resultativity. The *be* periphrasis, designated for stative resultativity, basically only allows COS verbs, as illustrated in the following comparison: *The door is closed*; *ice cream is melted*; *The window is broken*, etc. vs. **The door is knocked* (vs. *The door has been knocked*); **The car is driven* (vs. *The car has been driven*); **The man is hit* (vs. *The man has been hit*). This asymmetry between COS verbs and others must be attributed to their lexical semantics and associated with the inner aspect. Failing to capture this fact, the structure in (55) in turn dissociates the aspectual prefixes and inflectional status of the participle from the historical changes that

[36] McFadden & Alexiadou (2006: 276) write, 'The most important element here is the $\mathrm{Asp_R}$ head [in (55)]. It produces a state which is the result of a prior event, and thus requires as its complement an eventuality which can reasonably produce a result state. This explains why iteratives, duratives and atelic predicates in general don't like to show up in the *be* perfect.' Admittedly, their assumption seems to be on the right track. But this is far from sufficient to explain what happened to the participle itself.

have taken place in the meaning and function of the participle. This empirical difficulty exposes the analysis to a theoretical weakness.[37]

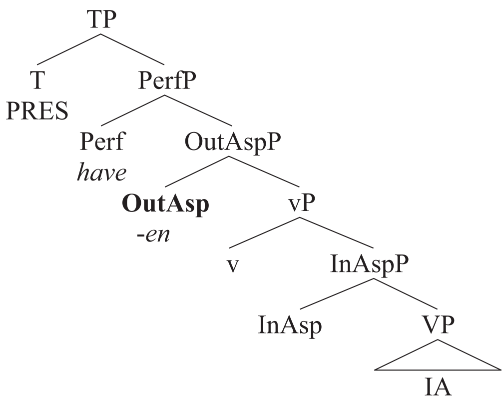
I would suggest that considerations should not be restricted to a (certain part of a) certain structure but should take a broader view covering all that may matter in a certain change and development. Thereafter, the changes occurred not only with the selection of heads, which are lexically realized as *have* and *be* in the case of clauses, but also with the formation of participles as demonstrated. The emergence of the perfect resultative sense in participles is related to the interaction between the outer and inner aspects, not merely to the outer aspect. Importantly, relevant developments took place with both prenominal and post-auxiliary participles.

5. Conclusion

In this article, we have seen that aspectual change has occurred in the formation of prenominal participles in the history of English. The prenominal participles were stative in OE and perfect in EME onwards. The locus of resultativity shifted from InAsp to OutAsp because of the loss of aspectual prefixes in ME. In particular, this has shifted the need to mark the required resultativity. As OutAsp became active, it brought about the perfect resultative sense as well as other dynamic/eventive meanings. However, stative resultatives have not disappeared because English has had verbs that inherently affect internal arguments without any morphemes.[38] Such verbs generally encode COS meanings lexically. Interestingly, throughout ME and EModE, numerous ergative verbs, which typically have COS meanings, emerged (see Bai 2016).

Finally, if the proposed analysis is correct, it compensates for the weakness of McFadden & Alexiadou's (2010) analysis. Tying the structures in (46)–(47) together with the structures of the *have* periphrasis and *be* periphrasis proposed by McFadden & Alexiadou (2010), we arrive at the analysis in (57)–(58).[39]

(57) The structure of the *have* periphrasis:



[37] McFadden (2015) investigates the prefix *ge-* and is aware of its function in participial formation. His analysis of this prefix compensates for the weakness of McFadden & Alexiadou's (2006, 2010) analysis of the *have* periphrasis and the *be* periphrasis to some extent.

[38] Stative resultatives have survived into PDE and presumably the majority of prenominal participles in PDE are still stative resultatives, e.g. *the closed door*, *the damaged car*, *the broken window*, etc.

[39] VoiceP is left out here, for the sake of simplicity.

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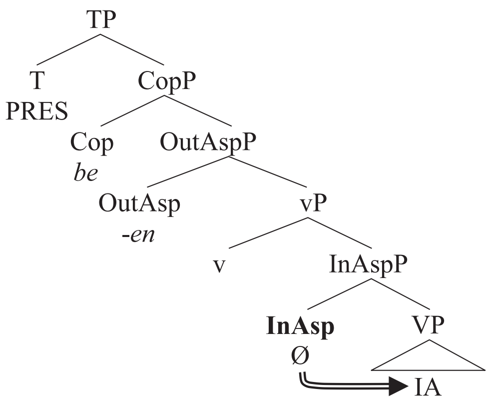
(58) The structure of the *be* periphrasis:



PerfP and CopP are projected above OutAspP in (57)–(58), respectively. For the *have* periphrasis, (57) is basically the same as in McFadden & Alexiadou's (2010) proposal (54)–(55). What is novel here for post-auxiliary participles (but not canonical simple pasts nor habituals) is the division of loci of resultativity. The locus is OutAsp in (57), which successfully captures the fact that the lexical semantics of the base are less constrained in perfect resultatives than in stative resultatives. The locus is InAsp rather than OutAsp in (58), which captures the fact that, normally, only COS verbs are input to stative resultatives, which are selected by copulas including *be* and others such as *remain* and *seem.* OutAspP, [+V] or [+Adj], is subject to subcategorization by a dominating head, Perf or Cop. It must be noted here that the structure in (58) has survived into PDE; it has not died out because of the loss of aspectual prefixes in EME.

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